

SMART WEARABLE SYSTEM FOR SEIZURE MONITORING AND EMERGENCY RESPONSE

A mobile application integrated with wearable sensors

By: Dana Almounayer & Jwdee Alamam

Instructor: Dr.Passent Alkafrawy

BUSINESS CANVA

KEY PARTNERS



(Clinical Partner)



(EEG data provider for research)



(Wearable technology platform)

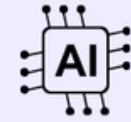
KEY ACTIVITIES

Implement rule-based detection logic

Process wearable sensor data

Maintain the mobile application

Test system accuracy and reliability



KEY RESOURCES

Simulated wearable sensor data

Mobile & backend development tools

Software developers

Cloud servers

VALUE PROPOSITIONS



Detects abnormal physiological patterns in real time



Provides early warnings and rapid emergency alerts



Helps prevent injuries and reduce anxiety



Non-invasive and wearable



Uses AI and real medical data (EEG)

CUSTOMER RELATIONSHIPS



Real-time alerts and notifications

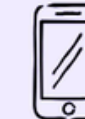
Mobile app support

Regular updates and improvements

Educational content about epilepsy safety

CHANNELS

Mobile application (Android / iOS)



Wearable device (smartwatch / wristband)

Hospitals and clinics
Online platforms



CUSTOMER SEGMENTS



People with epilepsy



Parents and caregivers of epilepsy patients



Hospitals and neurology clinics

COST STRUCTURE



Software development



Cloud infrastructure



Research and testing



Maintenance and updates

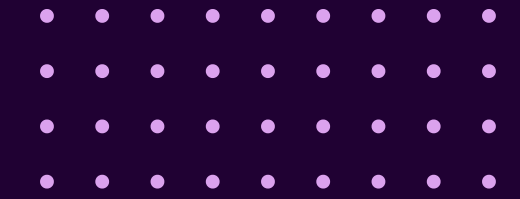
REVENUE STREAMS

Subscription for mobile app

Licensing to hospitals

Research collaboration funding

Government or health grants



PROBLEM OVERVIEW & RESEARCH GAPS

- Epilepsy is a neurological disorder affecting millions worldwide (World Health Organization, 2023)
- Seizures occur suddenly and can lead to injury or delayed emergency response (Fisher et al., 2014)
- Traditional monitoring relies on hospital-based EEG, which is costly and limited (Patel et al., 2012)
- Existing systems lack continuous real-time monitoring in daily life (Patel et al., 2012; Teixeira et al., 2011)
- Many solutions focus on detection but lack integrated alert and reporting systems (Koren et al., 2025; Darankoum et al., 2026)



SUSTAINABLE DEVELOPMENT GOALS (SDGs)

SDG Icon	Title	Description
	Good Health and Well-Being	 Improves patient safety through real-time seizure detection and faster emergency response.
	Industry, Innovation, and Infrastructure	 Uses AI and wearable technology to build an innovative and scalable healthcare system.
	Sustainable Cities and Communities	 Enhances safety and supports independent living for people with epilepsy.

COMPETITOR ANALYSIS

Empatica Embrace2

Accurate detection
No real time alerts

NeuroPace RNS

Effective clinically
Invasive (surgery required)

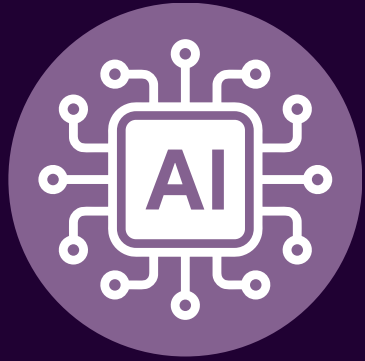
BrainBit EEG

Brain monitoring
No real-time alerts

Existing systems are limited and incomplete



PROPOSED SOLUTION



**AI-based seizure
detection using
EEG**



**Real-time
monitoring
system**



**Multi-sensor
integration (EEG +
heart + motion)**



**Mobile app for
users and
caregivers**

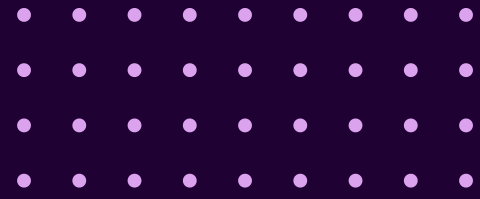


**Event logging
and
visualisations**



**Structured
reports for
doctors**

Purpose: Detecting Seizures in real time using AI and notify registered caregivers immediately to support patients, caregivers, and neurologists and improve safety, response time, and clinical decision making



SYSTEM REQUIREMENTS



Functional Requirements

- Detect seizures in real time using physiological sensor data
- Record the start and end time of each seizure & calc duration
- Store seizure frequency statistics
- Send immediate alerts when a seizure is detected and when seizures end
- Allow direct emergency contact calling
- Display seizure history in chronological order
- Provide visual charts
- filtering of seizure data by date
- Connect to a non-invasive wearable device
- Collect EEG, heart rate, and motion data & Sync data automatically to the mobile app

Non-Functional Requirements

- Send alerts within 5 seconds
- Ensure high reliability (99%)
- Support Android and iOS platforms
- Provide a simple and accessible interface
- Enable 24/7 continuous monitoring

***Key system requirements in brief**



ACTORS/PERSONAS



Patient

- Monitors real-time seizure data
- Reviews history & statistics



Caregiver

- Receives emergency alerts
- Responds quickly to seizures



Neurologist

- Reviews seizure analytics
- Generates medical reports



System

- Detects seizures & triggers alerts
- Logs and processes data

USE CASES

UC1 — Monitor Patient Status

View real-time patient data

UC2 — Manage Seizure Records

Log, review, and filter seizure records

UC3 — Emergency Alert & Response

Send alerts and support emergency action

UC4 — Clinical Review & Reports

View analytics and export reports

SYSTEM ARCHITECTURE

Presentation Layer

Mobile app interface

Application Layer

Backend logic and coordination

Data Processing Layer

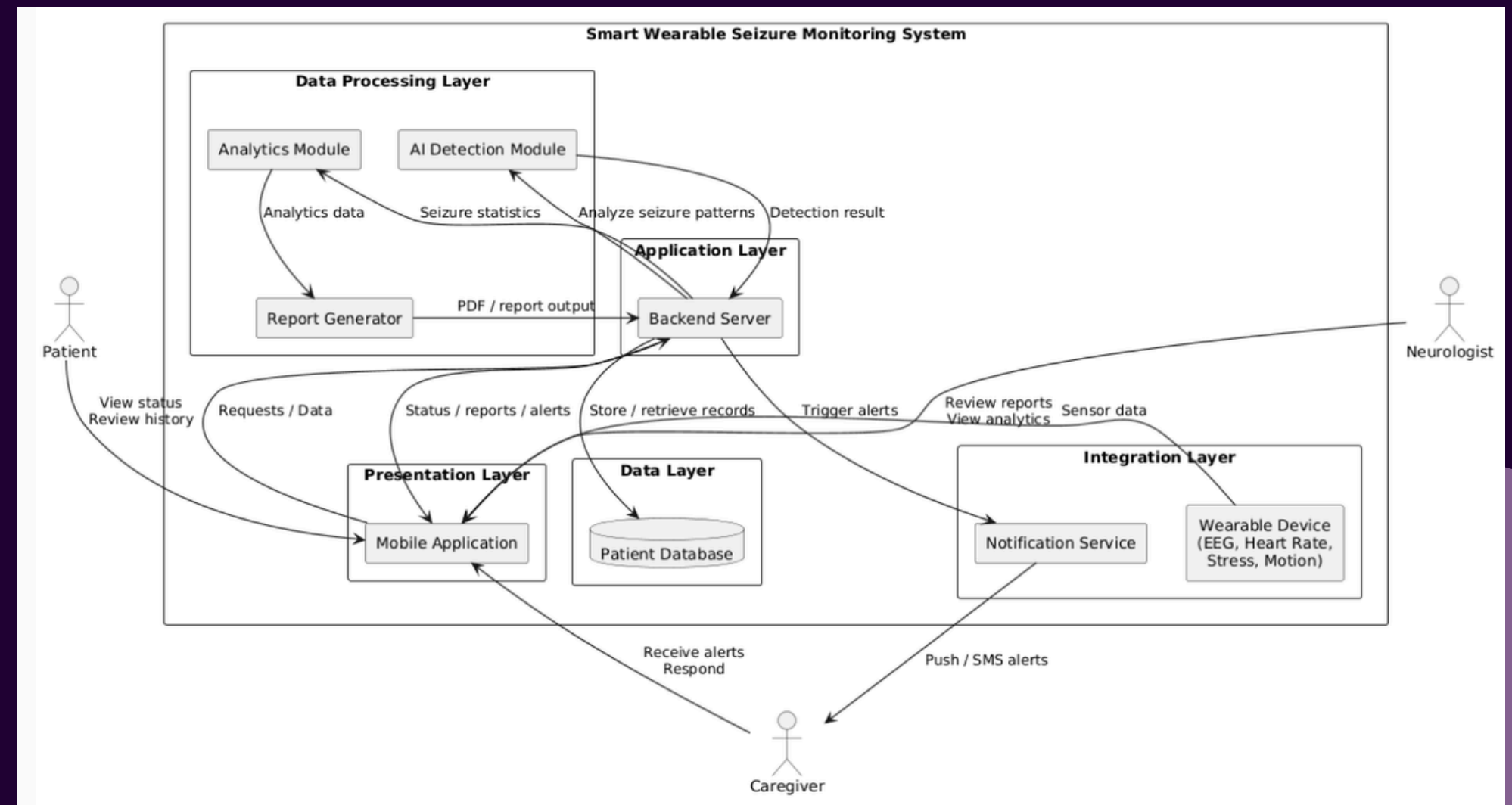
AI detection, analytics, reports

Data Layer

Patient and seizure database

Integration Layer

Wearable device and notification service



TECHNOLOGIES USED



**Backend & Cloud
System**



Database



Notification Service



**Cloud / Distributed
System**



**Secure
Communication**

SYSTEM WORKFLOW

**Data Collection &
Transmission**

Intelligent Analysis

**Real-time seizure
detection**

**Emergency
Response**

**Monitoring &
Reporting**



SYSTEM DESIGN

- The system follows a layered and modular design
- Each component has a specific responsibility
- The design improves organization, scalability, and maintainability



Observer Pattern

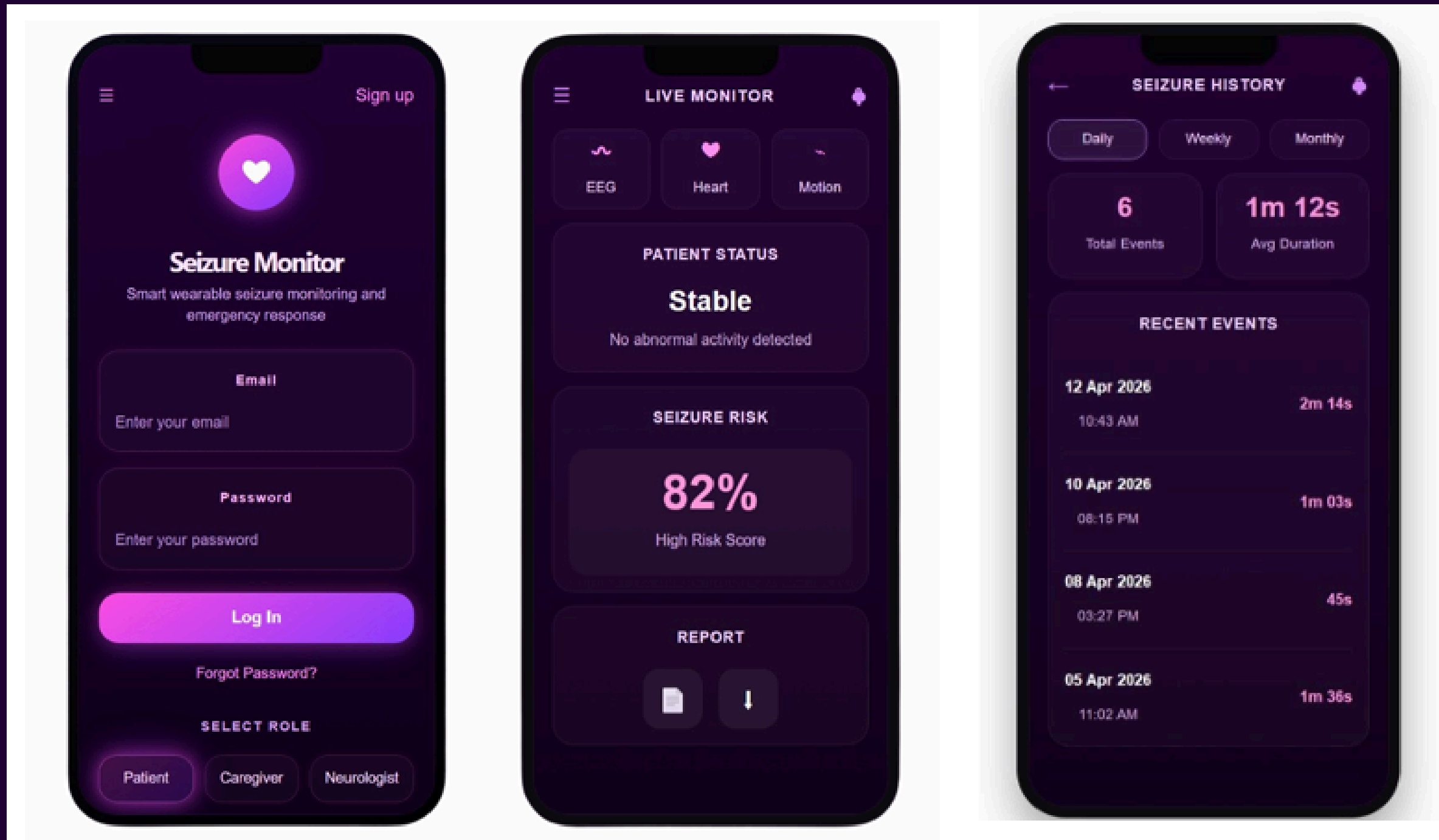
used for real-time notifications and alerts



MVC Pattern

separates interface, logic, and data handling

USER INTERFACE

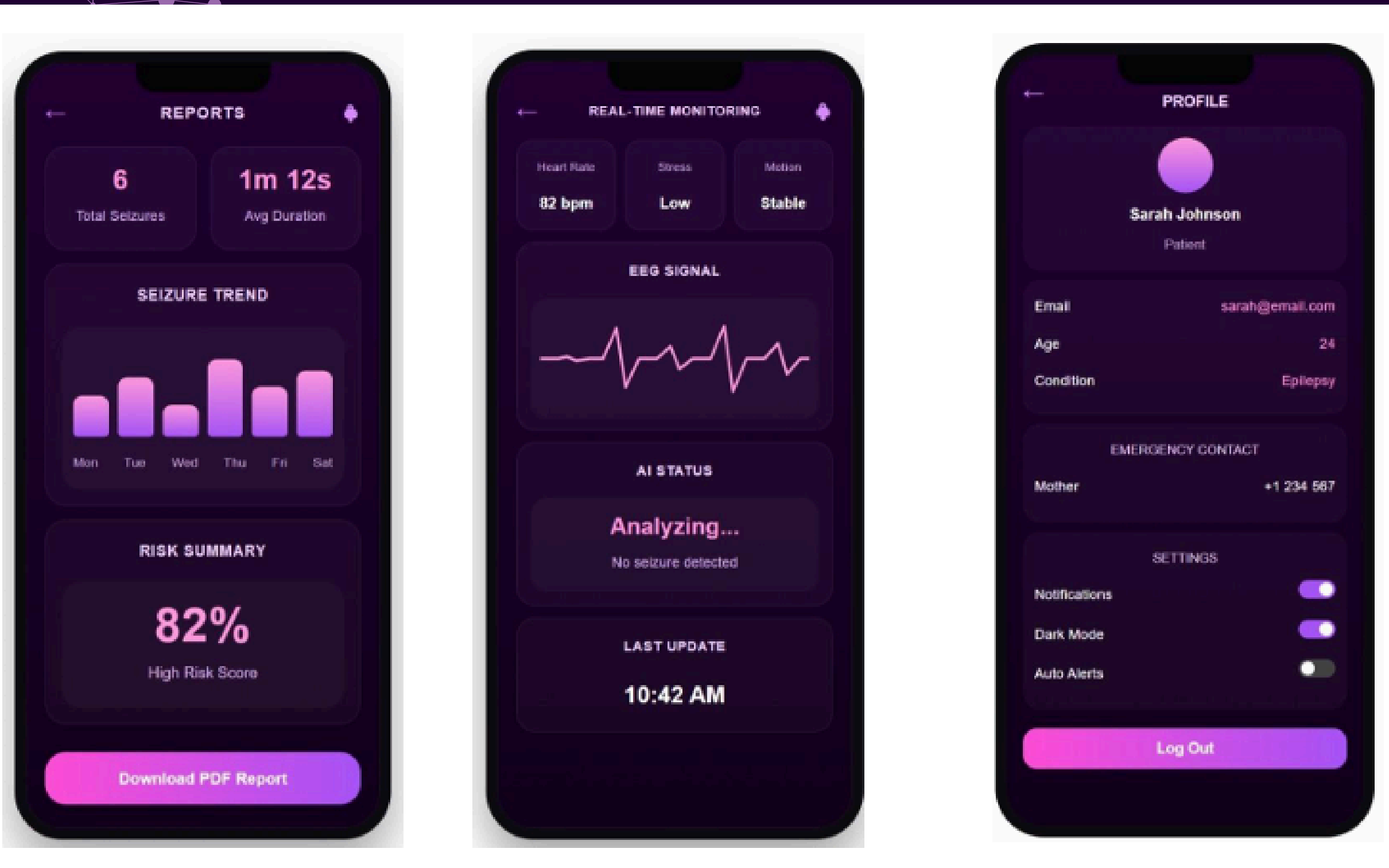


Simple layout

Easy navigation

clear alerts

Fast access



Simple layout

Easy navigation

clear alerts

Fast access

RISK MANAGEMENT

Identified risks and engineering mitigations

Risk	Impact	Mitigation
Sensor failure	High	Use reliable hardware and fallback mechanisms
Network failure	High	Offline buffering and SMS fallback delivery
Data privacy issues	High	AES-256 encryption and TLS 1.3 secure protocols
AI model inaccuracy	Medium	Continuous training, validation, and monitoring
System overload	Medium	Scalable cloud infrastructure with auto-scaling



CONCLUSION

A smarter, safer way to monitor epilepsy

Key Outcomes

- Improves patient safety through real-time monitoring
- Reduces emergency response time with instant alerts
- Empowers caregivers with mobile-centered tools
- Supports neurologists with structured analytics
- Combines wearables, AI, cloud, and secure design

References



Future Work



Smartwatch integration

expand to consumer wearables



GPS location tracking

pinpoint patient location during alerts



More health features

sleep, oxygen, and activity insights



THANKS FOR YOUR ATTENTION!